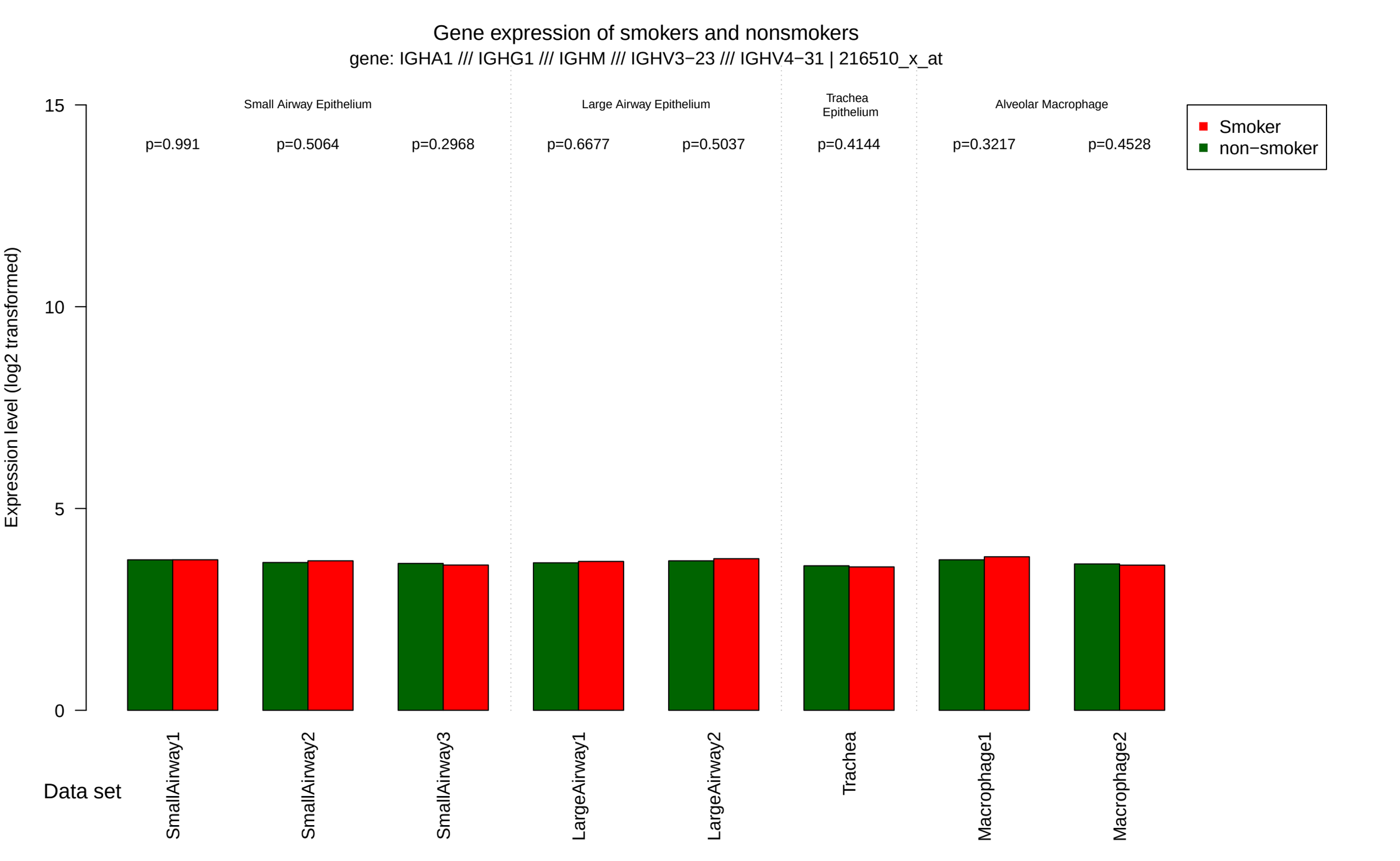
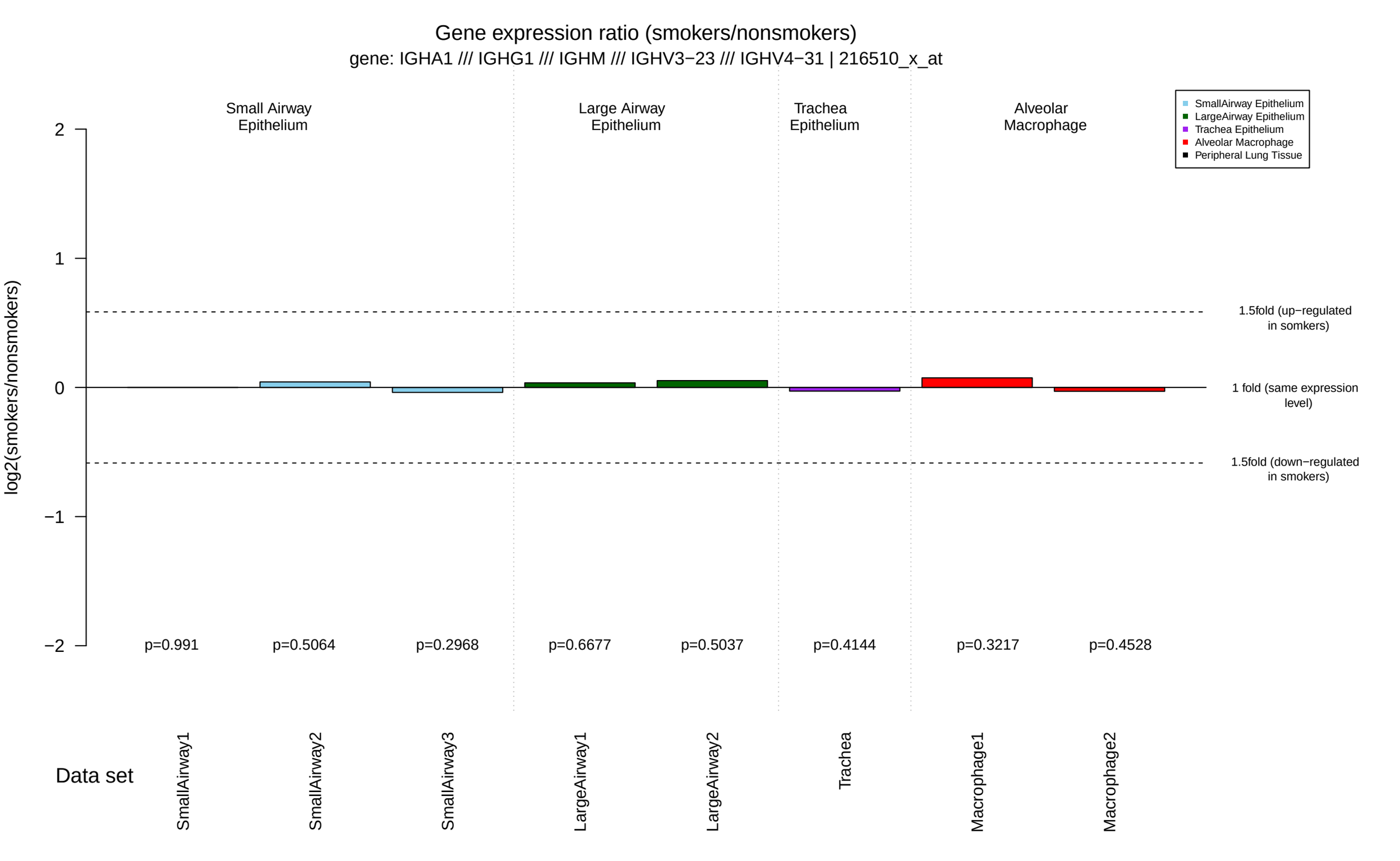
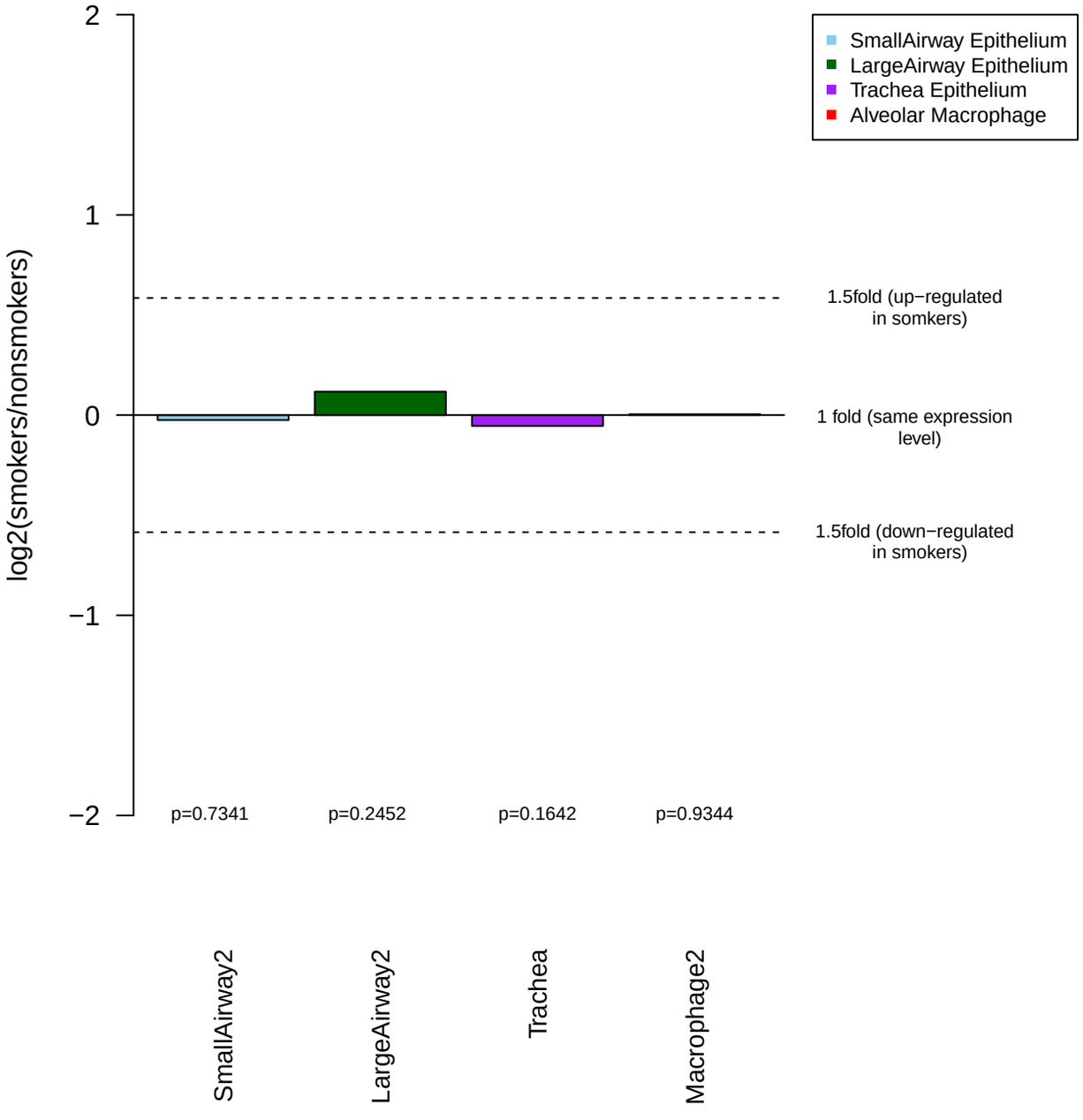

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.991
p=0.5064
p=0.2968
p=0.6677
p=0.5037
p=0.4144
p=0.3217
p=0.4528
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.991
p=0.5064
p=0.2968
p=0.6677
p=0.5037
p=0.4144
p=0.3217
p=0.4528
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

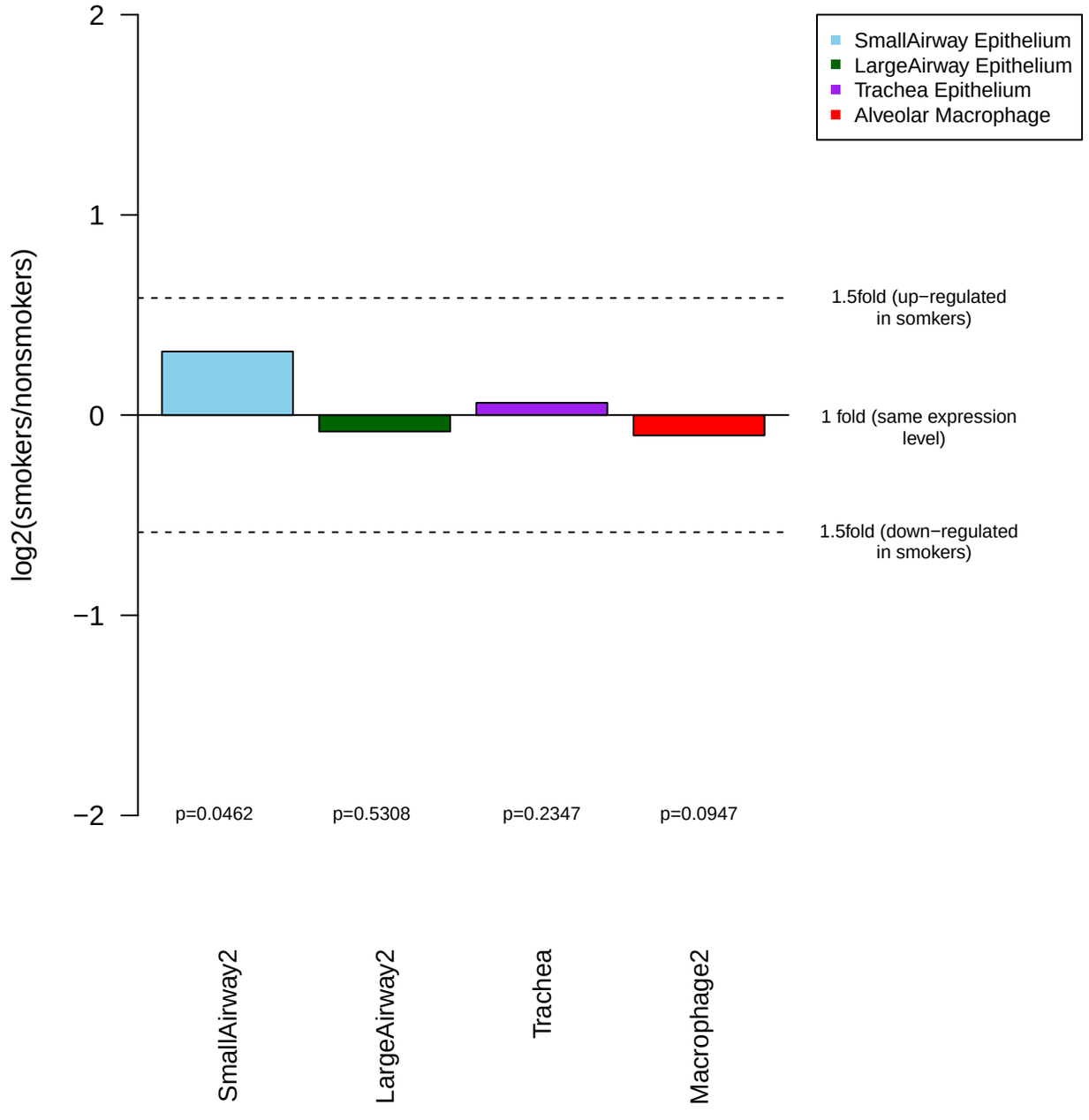
Gene expression ratio in Males

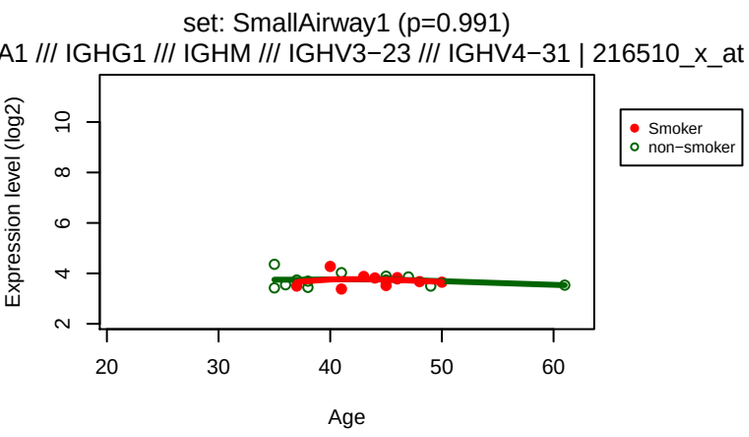
gene: IGHA1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at

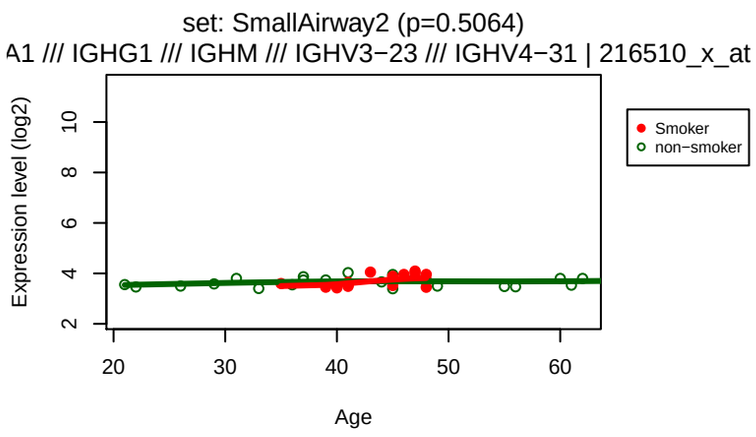


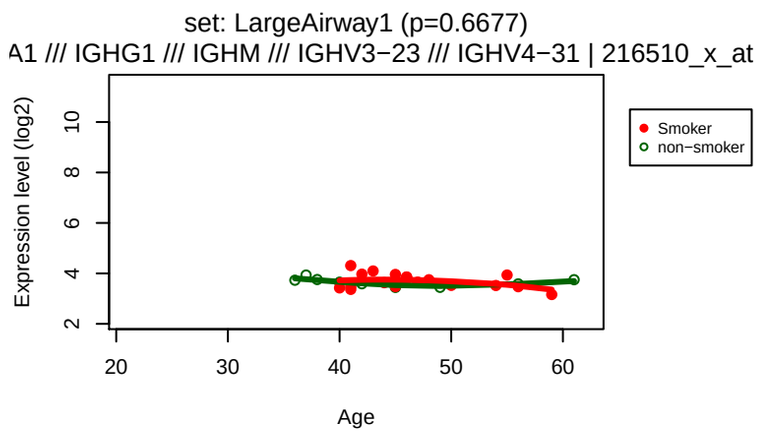
Gene expression ratio in Females

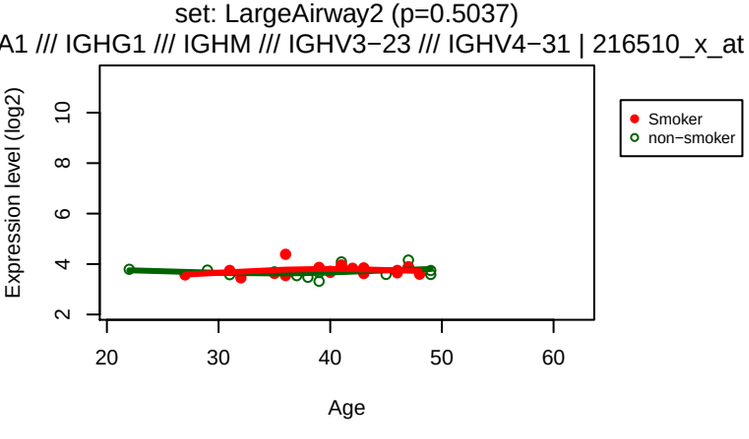
gene: IGHA1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at

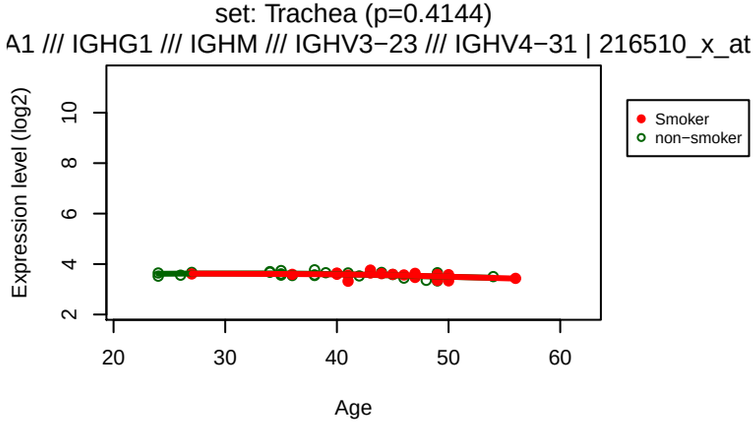


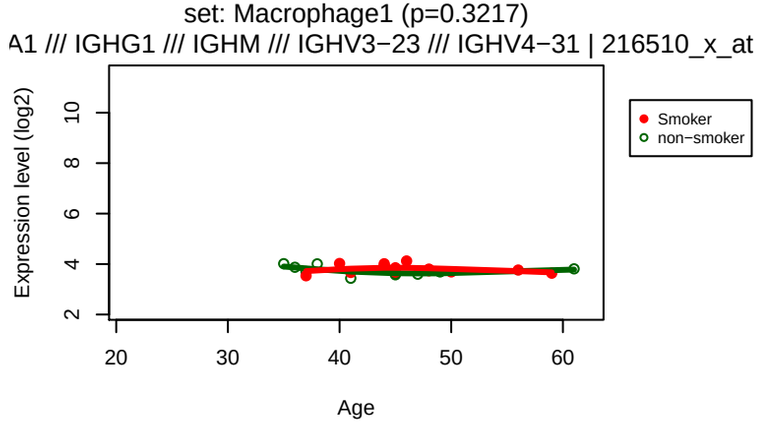

set: SmallAirway1 (p=0.991)
A1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Expression level (log2)
Age
Smoker
non−smoker

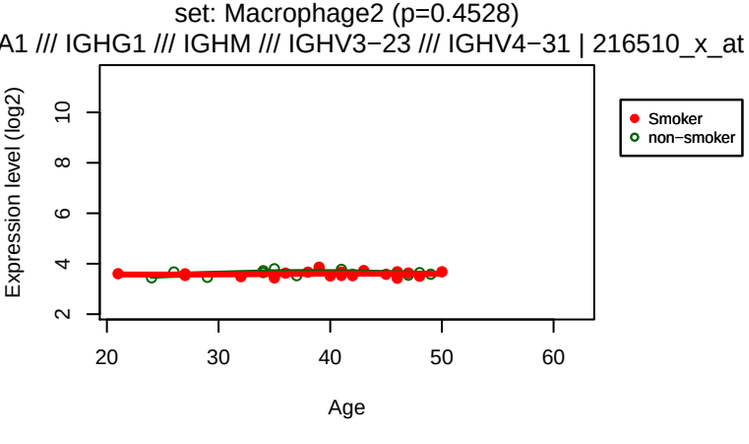

set: SmallAirway2 (p=0.5064)
A1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Expression level (log2)
Age
Smoker
non−smoker

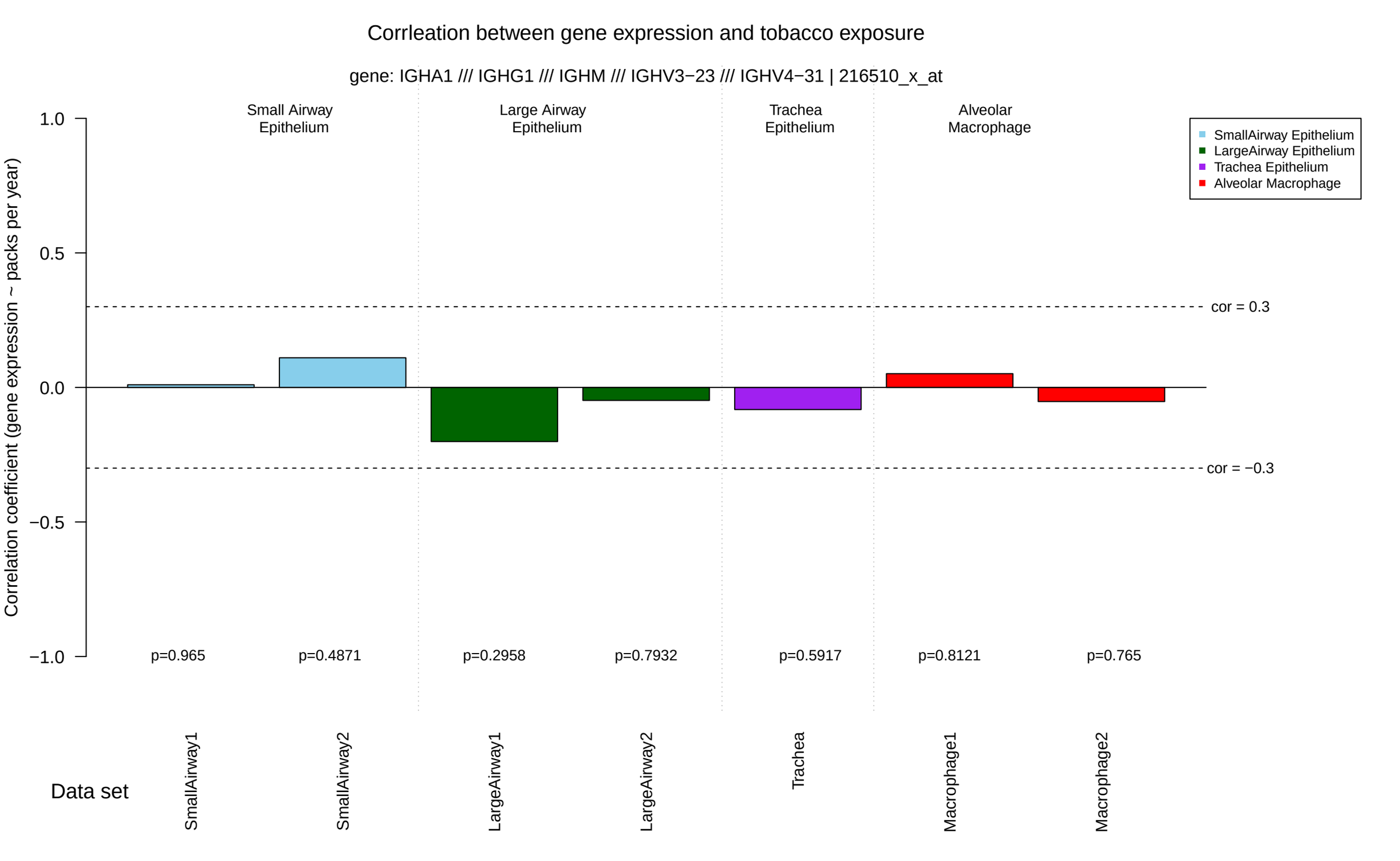

set: LargeAirway1 (p=0.6677)
A1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5037)
A1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4144)
A1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3217)
A1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4528)
A1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHG1 /// IGHM /// IGHV3−23 /// IGHV4−31 | 216510_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.965
p=0.4871
p=0.2958
p=0.7932
p=0.5917
p=0.8121
p=0.765
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2